

Practicum No.10**Date: 27/2/2026****Financial Feasibility – Case study****I. Practical Significance**

Financial feasibility assessment determines whether a startup idea is economically viable and sustainable. This practicum is designed to transition CSE, AIML, DSCS & DS students from conceptualizing a software-based idea to understanding its economic reality. For any tech start-up in domains like AI, data science, or software development, a financial feasibility report is a critical document that determines whether a business idea is economically viable and sustainable. This exercise immerses students in the financial planning process, teaching them to forecast costs, project revenues, and calculate key metrics like Break-Even Point (BEP) and Return on Investment (ROI). By creating this report for their own identified start-up, students develop a crucial entrepreneurial skill: the ability to scientifically evaluate and communicate the economic potential of a technical innovation. This bridges the gap between coding a solution and building a profitable business, preparing them for roles as tech founders, product managers, or innovation consultants who must justify project viability to stakeholders and investors.

This practicum exercise is expected to develop the following skills for the industry-identified competency.

- **Develop** a pitch deck, project proposals for small/medium scale startups logically and scientifically.
(As stated in the Course Structure)
- a) **Apply** financial principles to assess the viability of startup projects. (CDO)
- b) **Interpret** cost, revenue, and profitability indicators effectively. (CDO)
- c) **Exhibit** professional and ethical decision-making in financial planning. (ADO)

II. Relevant CO(s) (As stated in the Course Structure)

- **CO4. Design** a suitable business model and formulate financial plans for the selected product/service.

III. Practical Outcome (PrO) (As stated in the Course Structure)

- **Present** a financial feasibility report for **the identified** CSE, AIML, DSCS & DS start-up.

IV. Related ADO(s) (As stated in the Course Structure)

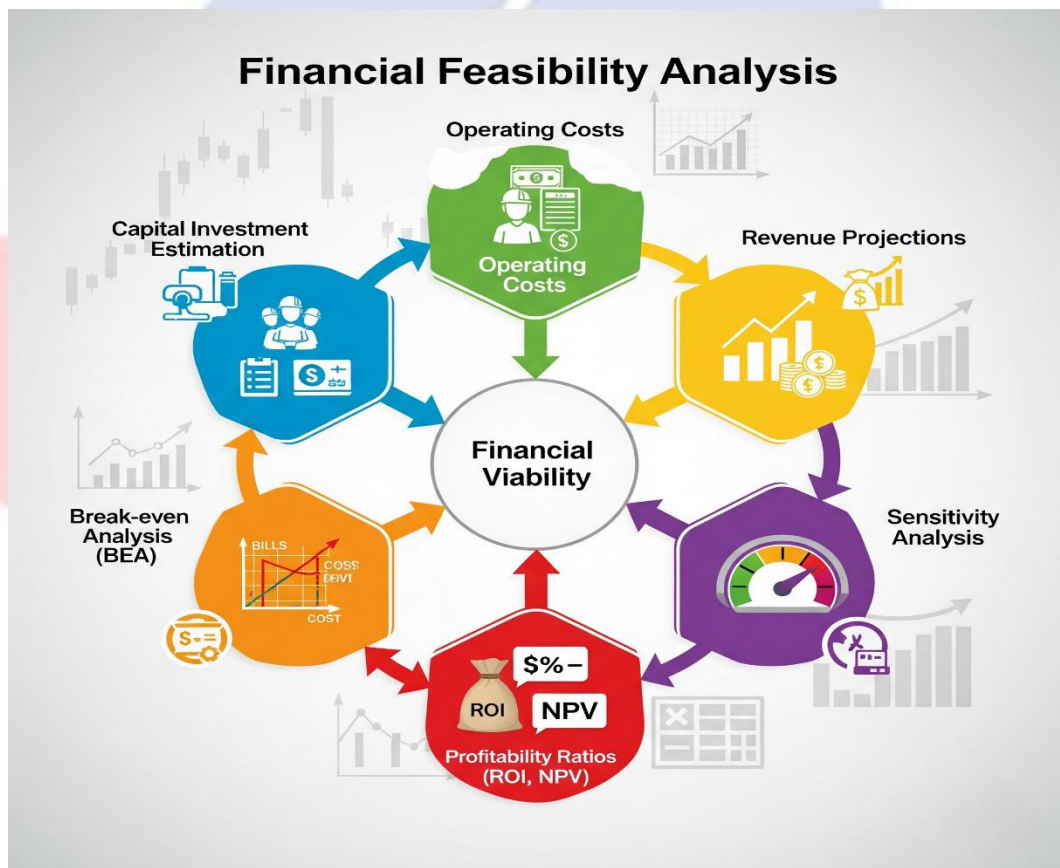
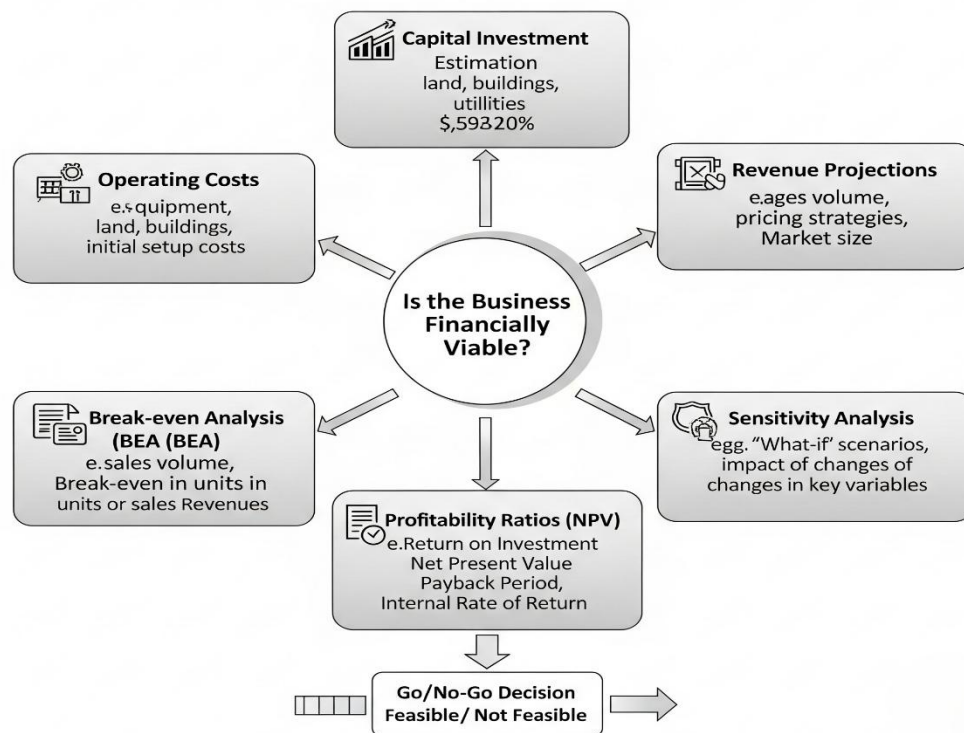
- a) **Function** effectively as a team member.
- b) **Take** responsible financial decisions aligned with entrepreneurial ethics.

V. Minimum Underpinning Theory (Include relevant content & images for this practicum.)

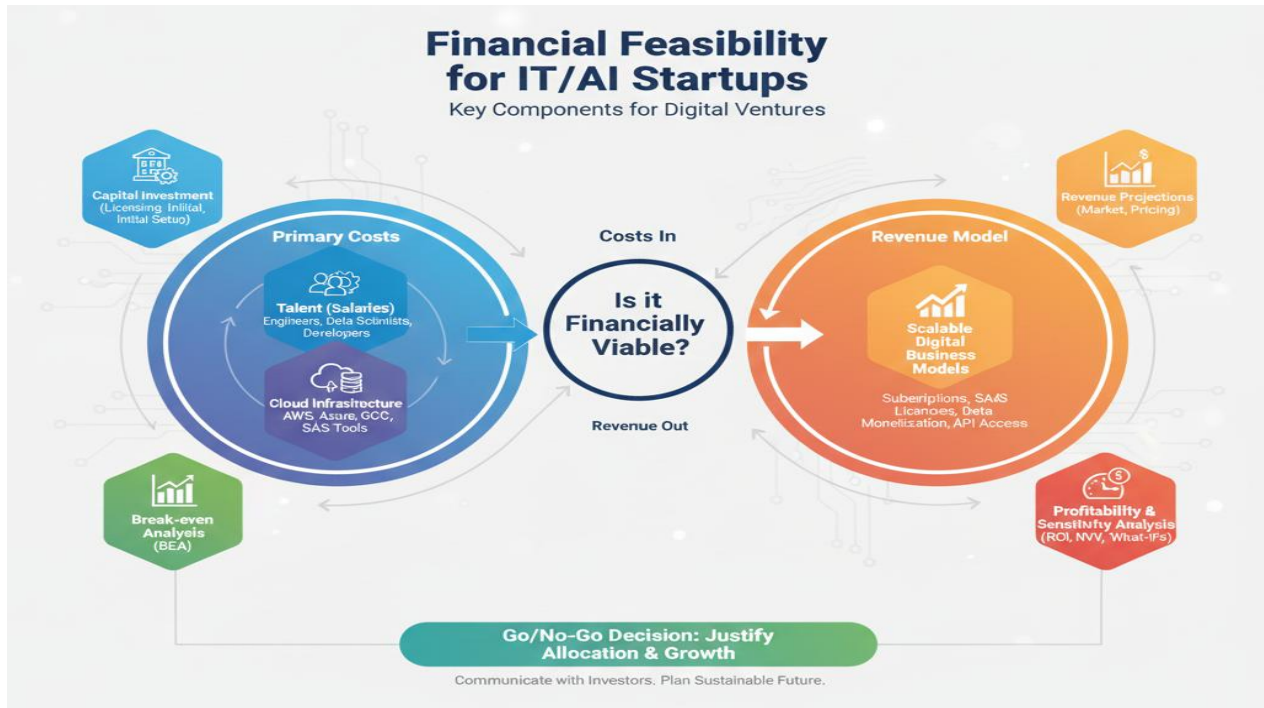
Financial feasibility involves estimating whether a business can generate sufficient revenue to cover its costs and yield profit within an acceptable timeframe. It typically includes the following components:

- Capital Investment Estimation – cost of equipment, materials, licensing, and infrastructure.
- Operating Costs – recurring expenses such as raw materials, utilities, labor, and maintenance.
- Revenue Projections – expected income based on market size, pricing, and sales volume.
- Break-even Analysis (BEA) – determining when total revenue equals total cost.
- Profitability Ratios – metrics like ROI (Return on Investment) and NPV (Net Present Value).
- Sensitivity Analysis – evaluating how variations in cost or price affect profitability.

6 Financial Feasibility Analysis



For software and AI start-ups, costs are often heavily weighted towards talent (salaries) and cloud infrastructure, while revenues are driven by scalable digital business models. Understanding financial feasibility equips engineering entrepreneurs to communicate effectively with investors, justify resource allocation, and plan sustainable growth.

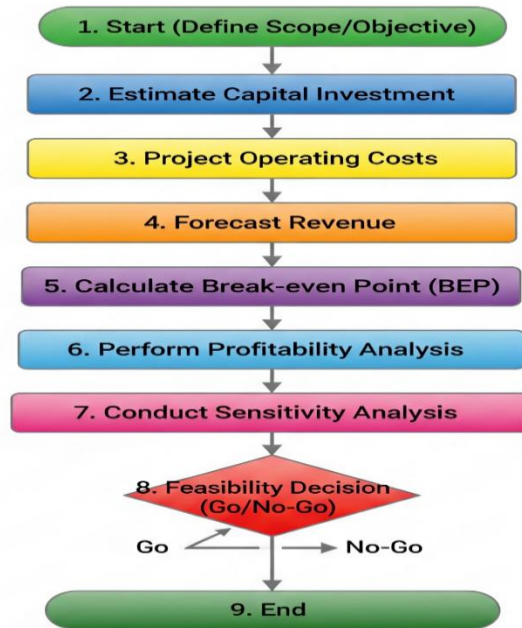


VI. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart (Include probable photos of experimental set-up.)

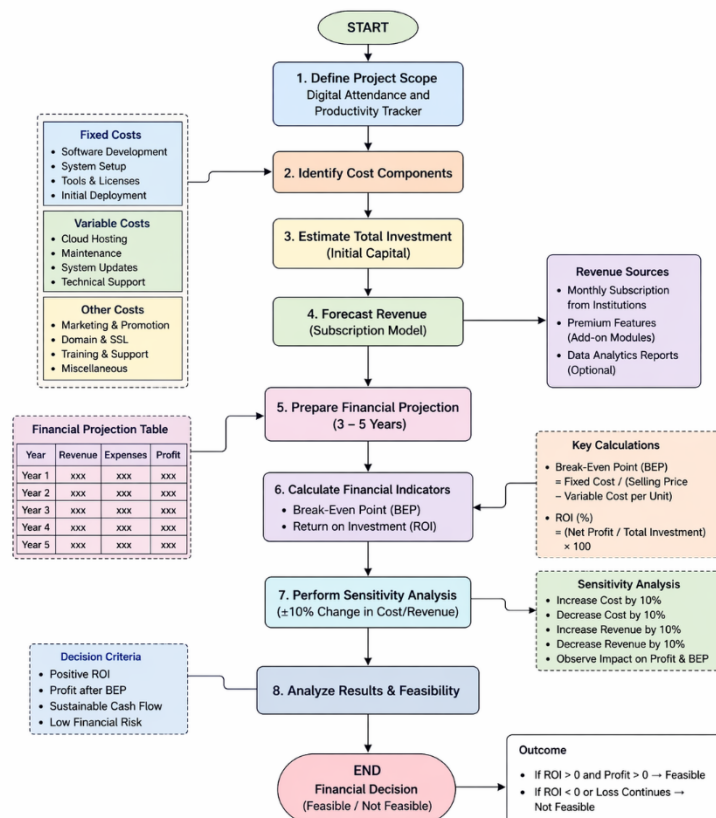
Algorithm Steps:

1. Start (Define Scope/Objective): Clearly state the financial goal being assessed (e.g., assessing viability over 3 years).
2. Estimate Capital Investment: Calculate initial, one-time costs, primarily for licensing, initial hardware/software, and essential setup fees.
3. Project Operating Costs: Focus heavily on the main recurring expenses for IT/AI: Talent/Salaries (e.g., data scientists, engineers) and Cloud Infrastructure (e.g., AWS, Azure, GCP costs).
4. Forecast Revenue: Base projections on a Scalable Digital Model (e.g., subscription tiers, SaaS licensing fees, data monetization).
5. Calculate Break-even Point (BEP): Determine the revenue or number of customers needed to cover all projected operating and capital costs.
6. Perform Profitability Analysis: Calculate key metrics like ROI (Return on Investment) and NPV (Net Present Value) to assess long-term value.
7. Conduct Sensitivity Analysis: Evaluate how profitability changes under different "what-if" scenarios, such as a 20% increase in talent wages or a 10% drop in subscription price.
8. Feasibility Decision (Go/No-Go): Based on the analysis, make a formal decision on whether the venture is financially sound.
9. End

Financial Feasibility Study Algorithm



FINANCIAL FEASIBILITY ANALYSIS PROCESS (Digital Attendance and Productivity Tracker)



Classroom or computer-lab based activity with spreadsheet tools (MS Excel / Google Sheets).

VII. Resources Required

S. No.	Resource	Suggested Broad Specification	Quantity
1	Laptop	i5 or above, MS Word / Google Sheets	1
2	Internet Access	For case study and cost estimation	shared
3	Calculator	For quick verification	shared
4	Sample Case Study	CSE, AIML, DSCS & DS -based startup	1

VIII. Safety Precautions

- Use realistic and justified financial data for calculations.
- Cite all sources used for cost or pricing estimates.
- Maintain accuracy and integrity in quantitative analysis.

IX. Procedure (Self-Instructional)

- Select** a case study of a technology-based startup (CSE, AIML, DSCS & DS domain).
- Study** the project background — product, target market, and technology scope.
- List** all cost components, including:
 - Fixed Costs (equipment, setup, R&D).
 - Variable Costs (materials, labor, logistics).
 - Overheads (marketing, maintenance, taxes).
- Estimate** total capital investment required for initial setup.
- Prepare** a projected income statement for 3–5 years showing revenue, expenses, and net profit.
- Calculate** the Break-Even Point (BEP) using:

$$\text{BEP (units)} = \frac{\text{Fixed Cost}}{\text{Selling Price per Unit} - \text{Variable Cost per Unit}}$$
- Compute** ROI or Payback Period to assess profitability.
- Conduct** sensitivity analysis by modifying cost or price assumptions ($\pm 10\%$) to observe impact.
- Interpret** results to determine whether the startup is financially feasible.
- Prepare** a Financial Feasibility Report (3–5 pages) including all calculations, graphs, and conclusions.

X. Actual Resources Used

S. No.	Name of Resource	Broad Specifications		Quantity	Remarks (If any)
		Make	Details		
1.	Laptop	ASUS	Used for financial analysis, documentation, and report preparation	1	-
2.	Internet Access	Wi-Fi	Used for researching startup costs, revenue models, and feasibility concepts	1	-
3.	Spreadsheet Software	Microsoft Excel	Used for BEP, ROI calculations and financial tables	1	-
4.	Reference Materials	Websites / Articles	Used for studying financial feasibility and startup case studies	1	-

XI. Actual Procedure Followed

1. Selected **Digital Attendance and Productivity Tracker** as the startup case study based on the mini project idea.
2. Identified the major **cost components** required for developing the system, including software development, cloud hosting, and database setup.
3. Estimated **fixed costs** such as application development, system setup, and initial deployment expenses.
4. Estimated **variable costs** including cloud server usage, system updates, and maintenance.
5. Defined a **revenue model** based on subscription fees charged to educational institutions for using the attendance and productivity tracking system.
6. Prepared a **financial projection table** estimating revenue, expenses, and profit over a period of 3–5 years.
7. Calculated key financial indicators including **Break-Even Point (BEP)** and **Return on Investment (ROI)**.
8. Performed a **sensitivity analysis** by adjusting revenue and cost values by $\pm 10\%$ to observe the impact on profitability.
9. Analysed the results to evaluate the **financial feasibility** of the Digital Attendance and Productivity Tracker startup idea.

XII. Observations

S. No.	Observations	Remarks
1	Initial investment estimated around ₹12,000	Includes development tools and initial hosting costs
2	Monthly operational cost around ₹3,000	Includes cloud hosting and system maintenance
3	Subscription fee estimated at ₹200 per institution per month	Affordable for schools and colleges

XIII. Results/Observations

1. The Digital Attendance and Productivity Tracker can be developed with a relatively low initial investment.
2. The subscription-based revenue model provides steady and recurring income from educational institutions.
3. The break-even point can be reached with a moderate number of subscribing institutions.
4. As more institutions adopt the system, profit increases steadily while operational costs remain stable.

XIV. Interpretation of Results

The financial analysis shows that the **Digital Attendance and Productivity Tracker** startup idea is financially feasible and scalable. The initial investment is relatively low, which reduces financial risk. The subscription-based revenue model ensures continuous income while the operational costs remain manageable. Achieving the break-even point with a reasonable number of users demonstrates that the business model is practical and sustainable.

XV. Conclusions

The financial feasibility study shows that the **Digital Attendance and Productivity Tracker** is a cost-effective and potentially profitable startup idea. The system requires moderate initial investment and can achieve break-even within a short time if adopted by educational institutions. With increasing adoption, the revenue can grow significantly. Therefore, the idea is financially viable and suitable for implementation in schools, colleges, and training institutions.

XVI. Suggested Practicum Related Questions:

Note: Below are a few questions related to industry-specific higher-order thinking skills (HOT) that teachers must use to ensure students achieve the predetermined course outcomes.

1. **Evaluate** how the break-even point reflects startup risk.
2. **Analyze** the relationship between pricing strategy and financial sustainability.
3. **Compare** different funding models (bootstrapping, venture capital, government grants).
4. **Propose** measures to reduce operating costs without compromising quality.
5. **Critique** how cost estimation inaccuracies affect decision-making.
6. **Design** a sensitivity test to analyze the effect of demand fluctuations on revenue.
7. **Justify** how realistic financial planning influences investor confidence.
8. **Predict** how scaling up production might alter the cost structure and profitability.

XVIII References / Suggestions for further reading

- a) MSME Startup Funding Handbook (2024–25).
- b) AICTE IDEA Lab Grant Guidelines.
- c) Startup India Resources – <https://www.startupindia.gov.in>.
- d) BIRAC & DST Funding Schemes (2024–25).

XIX Suggested Assessment Scheme

The performance indicators given serve as a guideline for assessing the ‘*process-related skills/LOs*’ (marks to be awarded in real-time in the laboratory by the faculty member, as these cannot be measured after the practicum is over) and ‘*product-related skills/LOs*’

Note: The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the competency related to the concerned course.

Performance Indicators		Maximum Marks	Marks Obtained
Process-related Skills: 18 Marks - 60%			
1	Collect realistic cost and revenue data.	6	
2	Work effectively as a team during case analysis.	5	
3	Apply correct formulas and reasoning in calculations.	4	
4	Demonstrate ethical and professional reporting.	3	
Product-related Skills: 12 Marks - 40%			
1	Complete detailed financial analysis with logical accuracy.	3	
2	Interpret results using graphs or tables effectively.	2	
3	Draw justified conclusions on financial feasibility.	3	
4	Conclusions.	2	
5	Practical related questions.	1	
6	Present the final report with clarity and format compliance.	1	
Total		30	

To be filled by Student:

S.No.	Name of the Student	Register No.
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1	Neavis Rishva. J	URK25CS1236
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To be filled by the Faculty Member:

Marks Obtained			Name and Designation of the Faculty Member	Date of Evaluation
Process-Related Skills (18)	Product-Related Skills (12)	Total (30)		

